

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE - RAIGAD - 402 103
Winter Semester Examination, December - 2017

Branch: B.Pharm.

Semester: I

Subject with Subject Code: Pharmaceutical Inorganic Chemistry
[BP104T]

Marks: 75

Date: 26 / 12 / 2017

Time: 3 Hrs.

Instructions:

- I. All questions are compulsory
- II. Illustrate your answers with neat sketches, diagram etc., wherever necessary.

Q.No.1 Multiple Choice Questions.

(20)

1. Ammonium chloride can be used as
 - a) Acidifier
 - b) Expectorant
 - c) Both acidifier & Expectorant
 - d) Emetic
2. Disinfectant which are used to maintain general public health are called as
 - a) Germicide
 - b) Antiseptic
 - c) Sanitizers
 - d) Bacteriostatic
3. What should not be the ideal property for an antacid preparation?
 - (i) Should be slow & short acting
 - (ii) Should not cause systemic alkalosis
 - (iii) Should buffer in the pH range 4-6
 - (iv) Should evolve gases
 - a) Only i
 - b) ii & iii
 - c) i & iv
 - d) Only iii
4. What is incorrect about sodium bicarbonate
 - a) It is systemic antacid
 - b) It also called as baking soda
 - c) It produce effervescence with acid
 - d) It does not absorb systemically
5. Fluoride inhibits caries formation via
 - (a) Decrease acid solubility of enamel
 - (b) bacterial inhibition
 - (c) Both a & b
 - (d) Increase acid solubility of enamel
6. Which of the following emetic is prepared by reaction of sodium bicarbonate with cream of tartar
 - a) Rochelle salt
 - b) Cupric sulphate
 - c) Epsom salt
 - d) None of the above
7. Which of the following antidote is act by adsorption
 - a) Sodium thiosulphate
 - b) Activated charcoal
 - c) Sodium nitrite
 - d) None of the above
8. Epsom salt is chemically
 - a) Ferrous sulphate
 - b) Magnesium sulphate
 - c) Ferrous gluconate
 - d) Copper sulphate
9. Which of the following is obtained from weathering volcanic ash
 - a) Kaolin
 - b) Bentonite
 - c) Aluminium hydroxide
 - d) Sodium orthophosphate

10. Identify the properties of potash alum

- i) Colorless crystal or powder, with slightly sweet taste
- ii) It act as astringent
- iii) Soluble in water & insoluble in ethanol
- iv) It gives positive tests for potassium, aluminium & sulphate

a) i, ii, & iv b) i, iii & iv c) ii, iii & iv d) all of the above

11) Indian Pharmacopoeia is published by _____

- a) Ministry of Health & Family Welfare
- b) Food & Drug Administration
- c) World Health Organization
- d) Ministry of Science & Technology

12) Citric acid is used in limit test of iron to prevent

- a) Precipitation of iron by ammonia
- b) Oxidation of iron
- c) Precipitation of iron by thioglycollic acid
- d) Complexation with thioglycollic acid

13) Barium sulphate reagent contains

- a) Barium chloride, sulphate free alcohol & sodium sulphate
- b) Barium chloride, sulphate free alcohol & Potassium sulphate
- c) Barium chloride, sulphate free alcohol, HCl & sodium sulphate
- d) Barium chloride, sulphate free alcohol, H_2SO_4 & sodium sulphate

12) Limit test of heavy metal is based on reaction between

- a) Metallic impurities & hydrogen sulphide
- b) Metallic impurity & Hydrogen sulphate
- b) Metallic impurities & acetic acid
- d) Metallic impurities & ammonia

15) Zinc oxide with eugenol is used in dental product as

- a) Dentifrices
- b) Medicated cement & fillers
- c) Polishing agent
- d) Anticaries agent

16) Lead acetate cotton plug is used in limit test of arsenic for

- a) Traps hydrogen sulphide
- b) Traps thiosulphate
- c) Traps hydrogen sulphate
- d) Traps arsine gas

17) Choose the correct composition of ORS

- (a) NaCl (b) KCl
- (c) Sodium citrate (d) sodium bicarbonate (e) glucose

(i) a, b, c, d

ii) a, b, d, e

iii) a, b, c, e

iv) b, c, d, e

18) Measurement of radioactivity is termed as

- (a) Radiation
- (b) Radiation dosimetry
- (c) Both (a) and (b)
- (d) None of the above

19) In which year the first Indian Pharmacopoeia was published by Indian Pharmacopoeia Committee?

- a) 1948
- b) 1960
- c) 1955
- d) 1966

- 20) Gamma rays are electromagnetic radiations with a wave length
- (a) Much larger than those of light
 - (b) Much shorter than those of light
 - (c) Equal to light
 - (d) None of the above

Q.No. 2 Attempt any TWO of the following:

(20)

- A) Define & classify antimicrobials. State the mechanism by which they act. Write any one methods of preparation & assay of hydrogen peroxide.
- B) Enlist sources of impurities. Draw a neat labeled diagram of Gutzeit Apparatus. Explain the principle of arsenic limit test.
- C) Define & classify antacids. Enlist its ideal properties & explain aluminium hydroxide gel in short.

Q.No. 3 Attempt any SEVEN of the following:

(35)

- A) Explain how buffer systems maintains physiological acid base balance.
- B) What are dentifrices? Explain Role of fluoride in the treatment of dental caries.
- C) Define antidote. Explain in short about sodium thiosulphate.
- D) Write the principle and reaction involved in limit test of iron.
- E) Write in short about acidifier with example.
- F) Define buffers. Write a note on buffer capacity.
- G) What are emetics? Explain copper sulphate with respect to properties & assay.
- H) Write a note on ORS.
- I) Enlist properties of α , β & γ particles. Write applications of radioisotopes.

----- END OF PAPER -----

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE - RAIGAD - 402 103
Supplementary Examination, June 2018**

Branch: B.Pharm

Semester: I

Subject with Subject Code: Pharmaceutical Inorganic Chemistry
[BP104T]

Marks: 75

Date: 08 / 06 / 2018

Time: 3 Hrs.

Instructions:

- I. All questions are compulsory.
- II. Candidates should begin the answer to each question on a new page.
- III. Illustrate your answers with neat sketches, diagram etc., wherever necessary.

Q. No. 1. Multiple choice questions:

(20)

[Each of the following questions has four alternatives. Only one of them is correct. Choose the correct answer.]

1. Ammonium chloride can be used as
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a) Kaolin b) Bentonite c) Aluminium hydroxide d) Sodium orthophosphate

10. Identify the properties of potash alum

- i) Colorless crystal or powder, with slightly sweet taste ii) It act as astringent
iii) Soluble in water & insoluble in ethanol iv) It gives positive tests for potassium, aluminium & sulphate
- a) i, ii, & iv b) i, iii & iv c) ii, iii & iv d) all of the above

11) Indian Pharmacopoeia is published by _____

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c) Metallic impurities & acetic acid
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c) Polishing agent d) Anticaries agent

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c) Traps hydrogen sulphate d) Traps arsine gas

17) Choose the correct composition of ORS

- (a) NaCl (b) KCl
(c) Sodium citrate (d) sodium bicarbonate (e) glucose
i) a, b, c, d ii) a, b, d, e iii) a, b, c, e iv) b, c, d, e

18) Measurement of radioactivity is termed as

- (a) Radiation (b) Radiation dosimetry
(c) Both (a) and (b) (d) None of the above

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20) Gamma rays are electromagnetic radiations with a wave length

- (a) Much larger than those of light
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Q. No. 2 Attempt any TWO of the following:

(20)

- a.) Define & classify antimicrobials. State the mechanism by which they act. Write any one methods of preparation & assay of hydrogen peroxide.
- b.) Enlist sources of impurities. Draw a neat labelled diagram of Gutzeit Apparatus. Explain the principle of arsenic limit test.
- c.) Define & classify antacids. Enlist its ideal properties & explain aluminium hydroxide gel in short.

Q. No. 3 Attempt any SEVEN of the following:

(35)

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- b.) What are dentifrices? Explain Role of fluoride in the treatment of dental caries.
- c.) Define antidote. Explain in short about sodium thiosulphate.
- d.) Write the principle and reaction involved in limit test of iron.
- e.) Write in short about acidifier with example.
- f.) Define buffers. Write a note on buffer capacity.
- g.) What are emetics? Explain copper sulphate with respect to properties & assay.
- h.) What are antidote? Write mechanism of action & assay of sodium thiosulphate.
- i.) Enlist properties of α , β & γ particles. Write applications of radioisotopes.



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Semester Examination – December 2018

Course: B. Pharmacy

Semester: I

Subject with Subject Code: Pharmaceutical Inorganic Chemistry (BP104T)

Date: 24/12/2018

Max Marks: 60

Time: 3 Hrs.

Instructions:

- I. All questions are compulsory.
- II. Candidates should begin the answer to each question on new page.
- III. Illustrate your answers with neat sketches, diagram etc; whichever necessary.

Q. No. 1. Multiple Choice Questions: (20)

(Each of the following questions has four alternatives. Only one of them is correct. Choose the correct answer.)

1. A mixture of weak acid with their salt of strong base is called as -----.
a) Acidic buffer b) Basic buffer c) Buffer d) All of these
2. The shelf life of commercial buffers is ----- years in unopened condition.
a) 1 b) 2 c) 3 d) 0.5
3. A solution having osmotic pressure more than blood plasma is called as ----- solution.
a) Isotonic b) Hypotonic c) Hypertonic d) None of these
4. ----- is the most abundant extracellular cation.
a) Potassium b) Calcium c) Sodium d) Magnesium
5. ----- microorganisms responsible for Dental Caries.
a) Streptococci b) Lactobacilli c) Both a & b d) None of these
6. ----- is clay composed of ash made from volcanos.
a) Kaolin b) Bentonite c) Gypsum d) Boric acid
7. The first edition of the Indian Pharmacopoeia was published in the year 1955 under the Chairmanship of -----.
a) Dr.R.N.Chopra b) Dr.B.N. Ghosh c) Dr.B.Mukherji d) Dr.Nityanand
8. Which method is used to purify the inorganic substances?
a) Washing b) Drying c) Sublimation d) All of them
9. The limit test for iron is based on the formation of pale pink to deep reddish purple color by reaction of iron with ----- acid.
a) Thioglycolic acid b) Nitric acid c) Acetic acid d) Hydrochloric acid
10. A monograph is complete description of a specific pharmaceutical which includes -----.
a) Chemical formula b) Molecular weight c) Solubility d) All of these
11. pH of the blood falls below 7.35 this condition is termed as -----.
a) Acidosis b) Alkalosis c) Achlorhydria d) Hyperchlorhydria
12. Unit of radioactivity is -----.
a) Calorie b) Newton c) Curie d) Joules



13. Synonym of aqueous iodine solution is -----.

- a) Lugol's solution
- b) Tincture iodine
- c) Strong iodine solution
- d) None of these

14. ----- is not a buffer system of our body.

- a) Borate buffer system
- b) Phosphate buffer system
- c) Carbonic acid-bicarbonate system
- d) Protein buffer system

15. Ammonium chloride is used as -----.

- a) Expectorant
- b) Systemic acidifier
- c) Diuretic
- d) All of these

16. ----- is only water soluble compound used as effective antacid.

- a) Magnesium trisilicate
- b) Milk of magnesia
- c) Sodium bicarbonate
- d) Sodium hydroxide

17. Assay of chlorinated lime is done by ----- titration method.

- a) Acid-base
- b) Complexometry
- c) Iodometry
- d) None of these

18. Anti microbial action of anti bacterial agents is due to -----.

- a) Protein precipitation
- b) Oxidation
- c) Halogenation
- d) All of these

19. Milk of magnesia I.P. is an aqueous suspension of hydrated -----.

- a) $MgSO_4$
- b) $Mg(OH)_2$
- c) $MgCO_3$
- d) Mg - trisilicate

20. The substances like..... which emit radioactive radiations are called as radioactive substances.

- a) Uranium
- b) Thorium
- c) Radium
- d) All of them

Q. No. 2. Attempt any TWO of the following:

(20)

- i) Define Radioactivity. Explain the properties of Alpha, Beta and Gamma rays. Write short note on storage and handling of radioactive material.
- ii) Write on – a) What are Astringents? Write the monograph of Zinc Sulphate.
b) Oral Rehydration Salt (ORS).
- iii) Define and classify Antacids with examples. Write the ideal properties of Antacids. Write the properties and uses of $NaHCO_3$.

Q. No. 3. Attempt any SEVEN of the following:

(35)

- a) What are Impurities? Explain different sources of impurities with examples.
- b) Explain in detail the assay of Ammonium chloride.
- c) Explain the role of fluoride in the treatment of Tooth Decay.
- d) Write the principle and reaction involved in limit test for Arsenic.
- e) Define buffer. Explain mechanism of buffer.
- f) Write in short history of Indian Pharmacopoeia.
- g) Define antimicrobial agents. Write the mechanism of actions of antimicrobials with examples.
- h) Write short note on Haematinics.
- i) Define and classify Antidotes with examples.

*** End ***



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Summer Semester Examination – May 2019

Course: B. Pharmacy

Semester: I

Subject with Subject Code: Pharmaceutical Inorganic Chemistry (BP104T)

Date: 21/05/2019

Max Marks: 75

Time: 3 Hrs.

Instructions:

I. All questions are compulsory.

II. Candidates should begin the answer to each question on new page.

III. Illustrate your answers with neat sketches, diagram etc; whichever necessary.

Q. No. 1. Multiple Choice Questions:

(20)

- Fifth Edition of Indian Pharmacopoeia was published in year?
A. 1995 B. 2007 C. 2008 D. 1996
- What is the role of Dil HNO₃ in limit test for chloride
A. To make solution Acidic B. To help formation AgCl ppt'd
C. Both A and B D. To make solution Neutral
- Which of the following Lewis base
A. NH₃ B. BF₃ C. AlCl₃ D. Fe⁺³
- 4 Isotonic sodium chloride solution contain
A. 0.7% w/v NaCl B. 1.1 % w/v NaCl C. 0.9% w/v NaCl D. 0.1% w/v NaCl
- Which of the following ion is major component of Extracellular Fluid?
A. Na⁺ B. Ca⁺² C. Mg⁺² D. K⁺
- Rock salt means
A. CaCO₃ B. KCl C. NaHCO₃ D. NaCl
- Which of the following is anticaries agent?
A. Sodium Fluoride B. Stannous fluoride C. Sodium monofluorophosphate D. All of above
- Which of the following is source of Eugenol
A. Vinca B. Clove C. Turmeric D. Ginger
- Antacid should be buffer in pH range
A. 1-3 B. 2-4 C. 4-6 D. 9-11
- Assay principle of Sodium bicarbonate is
A. Iodometric Titration B. Acid-Base Titration C. Precipitation Titration D. None of above
- Which of the following is an example of emollients?
A. Methyl cellulose B. Karaya Gum C. Liquid paraffin D. Isopgol
- Which of the following is not antimicrobial agent?
A. Antiseptic B. Antibacterial C. Anti-inflammatory D. Antifungal
- Which of the following is an example official agent which has anti microbial action?
A. Boric acid B. KMnO₄ C. H₂O₂ D. All of the above
- Eucalyptus is an example of
A. Acidifiers B. Expectorant C. Antacid D. Antidote
- Ammonium Chloride used as
A. Antacid B. Acidifiers C. Expectorant D. Both A and C

- END



DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY
LONERE – RAIGAD – 402 103

Winter Semester Examination- Winter – 2019

Branch: B. Pharm. (First Year)

Semester: I

Subject: Pharmaceutical Inorganic Chemistry

Subject code: BP104T

Marks: 75

Date: 20/12/2019

Time: 3 Hrs

- Instructions: i) All questions are compulsory.
ii) Figure to the right indicate full marks.
iii) Draw the diagram or flow charts wherever necessary.

Q. No. 1. Multiple Choice Questions (MCQs)

20 marks

Each of the following questions has four alternatives. Only one of them is correct. Choose the correct answer.

- 1) 0.9% W/V solution of sodium chloride contain.....mEq/L of sodium and chloride.
A) 273.50 B) 153.8 C) 136.75 D) 102.56
- 2) Fluoride inhibits caries formation via
A) Increase acid solubility of enamel B) Bacterial inhibition
C) Decrease acid solubility of enamel D) both B & C
- 3) The resistance of solution to change in PH upon addition of small amount of acid or alkali is termed as
A) Buffer action B) Buffer capacity C) Both a & b D) Buffer
- 4) is the synonym of ferrous sulphate?
A) White vitriol B) blue vitriol C) Green vitriol D) Muriatic acid
- 5) The principle function of astringent compound is ...
A) Contraction of capillary and small blood vessel by coagulation
B) Anti-inflammatory action by decreasing blood supply
C) anti-microbial action by protein precipitation
D) All of above
- 6) Ammonium chloride is used as.....
A) Expectorant B) Diuretics C) Acidifying agent D) All of the above
- 7) One of the following compounds is used as desensitizing agent.
A) Sodium fluoride B) calcium carbonate C) Zinc chloride D) Zinc eugenol cemen

- 8)is the principle anion of intra cellular fluid
- A) Phosphate B) bicarbonate C) sulphate D) Chloride
- 9) One of the following is correct for acid as per Lewis concept.
- A) Gives hydrogen ion in aqueous solution B) donate proton
- C) Accept proton D) Accept share in pair of electron
- 10) Following are the function of hydrochloric acid except
- A) Treat hyperacidity B) Soften fibrous food
- C) Activate proteolytic enzyme pepsin D) Kill bacteria in ingested food
- 11) The source of impurity in pharmaceuticals is due to
- A) Raw material B) Manufacturing process C) Improper storage
- D) All of the above
- 12) Which of the following class of antacid cause constipation as side effect
- A) Aluminium containing antacid B) Magnesium containing antacid
- C) Calcium containing antacid D) Both A & C
- 13) The group of GI agent commonly used to treat diarrhea is called as -----
- A) Adsorbent and protective B) Saline cathartics C) Both A&B D) antacid
- 14) Hydrogen peroxide acts as antimicrobial agent by ----- mechanism.
- A) Halogenations B) Oxidation C) Protein precipitation D) All of the above.
- 15) Epsom salt is synonym of ----- compound.
- A) Magnesium sulphate B) Sodium bicarbonate C) Calcium Carbonate D) Kaoline
- 16) -----is used to dissolve the precipitate of phosphate and carbonate impurity in limit test for chloride
- A) Barium chloride B) Nitric acid C) Silver nitrate D) All of above
- 17). Current edition of IP is published in-----
- A) 2017 B) 2018 C) 2019 D) 2016
- 18) Unit of radioactivity is-----
- A) Calorie B) Curie C) Joule D) Newton
- 19) Which of the following radiation have highest penetration power?
- A) Alfa B) Beta C) Gamma D) All of above

20) Radio Isotopes I^{131} used as

- A) Treatment of Hyperthyroidism
- B) Treatment of Hypothyroidism
- C) Diagnosis of thyroid gland
- D) all of the above

Q. No. 2. Long Answers (Answer 2 out of 3)

20 marks

- A) Define and classify antidote with suitable example. Explain properties, uses and mechanism of action of physiological and chemical antidotes used in cyanide poisoning.
- B) What is purity of Pharmaceuticals? Explain in detail sources of impurity in pharmaceuticals with suitable examples.
- C) What are radiopharmaceuticals? Give therapeutic and diagnostic application of Radiopharmaceuticals.

Q. No. 3. Short Answers (Answer 7 out of 9)

35Marks

- A) Explain the Principle and apparatus for the Limit test for Arsenic.
- B) Discuss synonym, method of preparation, properties, assay and uses of sodium bicarbonate
- C) What are haematinics? Give method of preparation, properties and medicinal uses of ferrous Sulphate.
- D) What are Anticaries agents? Explain role of fluoride in tooth decay.
- E) Write a note on astringent.
- F) What are expectorant? Explain properties, method of preparation, assay and uses of Ammonium Chloride.
- G) Define and classify topical agents. Write mechanism of antimicrobial agents with examples.
- H) Write a note on Physiological acid base balance
- I) What are buffer? Write Importance of Buffer solution in Pharmacy

Paper End

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Examination – Winter 2022

Date: 08/05/2023

Course : B. Pharmacy
Subject Name : Pharmaceutical Inorganic Chemistry
Max Marks : 75

Sem : I
Subject Code : BP104T
Duration : 3 Hr.

Instructions:

- All questions are compulsory
- Draw diagrams / figures wherever necessary
- Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions)

(10 x 2) = 20

- Define Emetics? Give two examples.
- Write down the properties and uses of Boric acid.
- Write down the properties and uses of Ammonium Chloride.
- Define the terms. i) Acidic Buffer ii) Buffer capacity.
- What are Radiopharmaceuticals?
- Define Expectorant. Give two examples.
- Draw a well labeled diagram of Gutzeit apparatus.
- Define Cathartics with examples.
- Write down reaction involved in limit test for Chloride.
- Classify Gastrointestinal agents with examples.

Q. 2. Long Answers (Answer 2 out of 3)

(2 x 10) = 20

- What are Antimicrobial agents? Explain the mechanism of actions of antimicrobial agents with suitable examples. Add a note on Hydrogen Peroxide.
- What is impurity? Enlist and explain sources of impurities in the pharmaceutical substances with examples.
- Write down the role of major Extra and Intra cellular electrolytes with suitable examples. Give the composition of ORS.

Q. 3. Short Answers (Answer 7 out of 9)

(7 x 5) = 35

- Write down the principle, reaction and procedure involved in the limit test of Iron.

- ii) Define antidote. Give classification of antidote. Discuss properties and uses of Sodium thiosulphate.
- iii) Classify Antacids with suitable examples. Add a note on combination of antacids.
- iv) Add a note on history of I.P.
- v) Write down the various applications of Radioactive substances.
- vi) Define Haematinics. Give synonym, properties, method of preparation and uses of FeSO_4 .
- vii) Explain buffer action and buffer equation.
- viii) Write short note on astringents.
- ix) What are Anticaries agents? Explain the role of fluoride ions in the treatment of Tooth Decay.

-----END OF THE PAPER-----

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

End Semester Supplementary Examination – Summer 2023

Date: 14-09-2023

Course : B. Pharmacy
Subject Name : Pharmaceutical Inorganic Chemistry
Max Marks : 75

Sem: I
Subject Code BP104T
Duration : 3 Hr.

Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Define Achlorhydria and give its treatment.
- ii) Define astringents with example.
- iii) Define Dentifrices and give its example.
- iv) Write the reaction involved in the limit test of sulphate.
- v) Define antiseptic and disinfectant.
- vi) Give the chemical formula and medicinal uses of sodium thiosulphate.
- vii) What is the alpha particle decay and give its suitable example?
- viii) Give the preparation and uses of CuSO_4 .
- ix) Give the composition of ORS.
- x) Define hyponatremia and hypernatremia.

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Define impurities and describe the various sources of impurities present in pharmaceutical substances with examples.
- ii) Define Buffers. Explain the mechanism of buffer action with an example. Briefly discuss the role of Buffers in Pharmacy.
- iii) Describe in detail Geiger - Muller counters and Scintillation counters for the measurement of Radio - activity.

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Define limit test and write down the principle, reaction, procedure involved in the limit test of chloride.
- ii) Write a note on cyanide poisoning.
- iii) Define and Classify antacids with suitable example and give the ideal properties of Antacid.
- iv) Write molecular formula, properties, method of preparation, assay and uses of

sodium chloride.

- v) Define Antimicrobial agent. Discuss mechanism of action (MOA) of inorganic antimicrobial agent.
- vi) Define and classify expectorant with its mechanism of action (MOA).
- vii) Define anticaries agents. Explain how fluoride produces anti-caries activity.
- viii) Name the disease caused by iron deficiency? Explain role of iron in the body.
- ix) Write the pharmaceutical application of radioactive substance.

-----END OF THE PAPER-----

End Semester Examination – Winter 2023

Date: 08/01/2024

Course : F.Y. B. Pharmacy

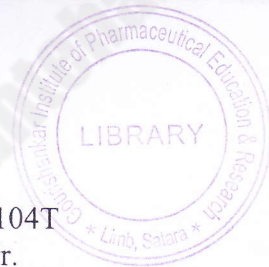
Sem : I

Subject Name : Pharmaceutical Inorganic Chemistry

Subject Code : BP104T

Max Marks : 75

Duration : 3 Hr.



Instructions:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q. 1. Objective Type Questions (Answer all the questions) (10 x 2) = 20

- i) Write principle involved in limit test for Chloride.
- ii) Define the term a) Hypernatremia, b) Hygroscopic
- iii) Define Pharmacopoeia with example.
- iv) Define hematinic with example
- v) Why combination antacids are given?
- vi) Define emetics with example.
- vii) What is half-life of radioisotope
- viii) Enlist two properties of α -ray.
- ix) Classify pharmaceutical buffers.
- x) Write synonyms for ferrous Sulphate & Magnesium Sulphate

Q. 2. Long Answers (Answer 2 out of 3) (2 x 10) = 20

- i) Define & classify impurities. Discuss various sources of impurities in pharmaceutical substances with suitable examples.
- ii) Define and give classification of gastrointestinal tract agents. What are antacids? Give ideal properties of antacids. Write preparation, properties uses and assay method for Sodium Bicarbonate.
- iii) Define and classify antidote with suitable example. Discuss their mechanism of action. Write preparation, properties, uses, assay method and mechanism of action of chemical antidotes used in cyanide poisoning

Q. 3. Short Answers (Answer 7 out of 9) (7 x 5) = 35

- i) Write short note on Expectorants.
- ii) Write short note on anti-caries agents.
- iii) Discuss the role of major extra & intra cellular electrolytes
- iv) What are antimicrobial agents? Discuss their mechanism of action.
- v) Write a note on limit test for arsenic.
- vi) What are astringents? Give their mechanism of action. Write properties & uses of potash alum.
- vii) What are buffers? Write various applications of buffers.
- viii) What are Radiopharmaceuticals? Discuss storage conditions, precautions & handling of radioactive substances.
- ix) Write a note on history of Indian Pharmacopoeia

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Course: B. Pharmacy

Supplementary Summer Examination – 2024

Subject Name: Pharmaceutical Inorganic Chemistry

Semester: I

Max Marks: 75

Date: 21-06-2024

Subject Code: BP104T

Duration: 3 Hr.

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams/figures wherever necessary
3. Figures to the right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions)

(10 X 2) = 20

- i) Explain the terms 'pharmacopeia' and 'monograph'
- ii) Define anti-carries agents with examples
- iii) Write the principle involved in the limit test for Chloride
- iv) Why are combination antacids given?
- v) Define Hematinics with an example.
- vi) What is the half-life of radioisotope
- vii) Define Antiseptics & Disinfectant
- viii) Write uses of 1. Potassium Permanganate 2. Sodium thiosulphate
- ix) Define Acid and base as per various theories.
- x) Classify cathartics.

Q.2. Long Answers (Answer 2 out of 3)

(10 X 2) = 20

- i) Define and give the classification of gastrointestinal tract agents. What are antacids? Give the ideal properties of antacids. Write preparation, properties used, and assay method for Sodium Bicarbonate.
- ii) What are intra and extra-cellular ions? Give their important physiological roles & related disorders.
- iii) Define and classify antimicrobial agents. Discuss their mechanism of action. Write preparation, properties, uses, and assay method for Hydrogen peroxide

Q.3. Short Answers (Answer 7 out of 9)

(5 X 7) = 35

- i) Define and classify antidotes with suitable examples. Discuss their mechanism of action.
- ii) Write a note on the history of Indian Pharmacopoeia
- iii) What are radiopharmaceuticals? Write properties of α , β , γ radiations
- iv) Write a short note on expectorants.
- v) What are astringents? Give their mechanism of action. Write properties & uses of potash alum.
- vi) Write a note on Electrolyte combination therapy.
- vii) Define Emetics. Write preparation and properties, uses of Copper Sulphate (CuSO_4)
- viii) Define and classify different types of impurities
- ix) What are buffer, and buffer capacity? Give types of buffers with their applications.

*** END OF THE PAPER ***